MTH 132(C and K) Dr. James Ashe
Exam 3 Tuesday 1st April, 2014 **Solutions**

You must show work to receive credit.

Find the simple interest AND the future value of the loan.

1. \$1000 at 4.25% for 6 months.

FV: $1000\left(1 + .0425 \cdot \frac{6}{12}\right) = \$1021.25$

interest: $1021.25 - 1000 = \$21.25$

Find the compound amount for the given deposit and the amount of interest earned.

2. \$2500 at 6.01% compounded monthly for 38 months.

FV: $2500\left(1 + \frac{.0601}{12}\right)^{12 \cdot \frac{38}{12}} \approx \$3022.65$

interest: $3022.65 - 2500 = 522.65$

3. \$2500 at 7% compounded continuously for 5 years.

FV: $2500e^{(.07 \cdot 5)} \approx \$3547.67$

interest: $3547.67 - 2500 = \$1047.67$

Find the amount that should be invested now to accumulate the following amount, if the money is compounded as indicated.

4. \$6200 at 9.5% compounded quarterly for 29 months.

$6200(1 + \frac{.095}{4})^{-(4 \cdot \frac{29}{12})} \approx \$4941.41$

Solve

5. Jane needs \$500,000 in 25 years to retire. To meet this goal she will invest money into an account paying 7% annual interest rate compounded continuously. How much money needs to be invested into the account?

$500,000e^{-(25 \cdot .07)} \approx \$86,886.97$

6. You place \$1500 into an account which pays 5% compounded daily. How much money will be in the account after 800 days? Assume that there are 360 days in a year and 30 days in a month.

$1500(1 + \frac{.05}{360})^{\left(360 \cdot \frac{800}{360}\right)} \approx \$1676.27$

Find the future value of each ordinary annuity, if payments are made and interest is compounded as given. Then determine how much of this value is from interest.

7. $R = \$3,250$ (payments); 11% interest compounded semi-annually for 7 years.

N:	$2 \cdot 7 = 14$	
`I%:`	11	
`PV:`	0	
`PMT:`	-3250	
`FV:`	?	$\Rightarrow$`FV` :\$65,950.86
`P/Y:`	2	
`C/Y:`	2	

$$S = 3250\left[\frac{(1+{.11}/{2})^{2\cdot 7}-1}{{.11}/{2}}\right] \approx \$65,950.86$$

and $65950.86 - 3250 \cdot 14 = \$20,450.86$ is from interest.

Find the periodic payment that will amount to each given sum under the given conditions.

8. $S = \$100,000$; interest is 5% compounded annually; payments are made at the end of each year for 12 years.

N:	$12 \cdot 1$	
`I%:`	5	
`PV:`	0	
`PMT:`	?	$\Rightarrow$`PMT` : -6282.5410
`FV:`	100,000	so payments of \$6282.54.
`P/Y:`	1	
`C/Y:`	1	

$$R = \frac{100000\,({.05}/{1})}{(1+{,05}/{1})^{12\cdot 1}-1} \approx \$6,282.54$$

Find the future value of each ordinary annuity, if payments are made and interest is compounded as given. Then determine how much of this value is from interest.

9. Payments of \$800 into an account paying 6.51% interest compounded quarterly for 12 years.

N:	$4 \cdot 12 = 48$	
`I%:`	6.51	
`PV:`	0	
`PMT:`	-800	
`FV:`	?	$\Rightarrow$`FV` :\$57,531.14.
`P/Y:`	4	
`C/Y:`	4	

$$S = 800\left[\frac{(1+{.0651}/{4})^{4\cdot 12}-1}{{..0651}/{4}}\right] \approx \$57,531.14$$

Total payments were $800 \cdot 48 = 38,400$. So $57,531.14 - 38,400 = \$19,131.14$ is from interest.

Find the present value of each ordinary annuity.

10. Payments of \$100 each month for 10 years at 8.5%.

N:	$12 \cdot 10 = 120$	
`I%:`	8.5	
`PV:`	?	
`PMT:`	-100	$\Rightarrow$ `PV` :\$8065.45
`FV:`	0	
`P/Y:`	12	
`C/Y:`	12	

$$P = 100\left[\frac{1-(1+{.085}/{12})^{-10\cdot 12}}{{.085}/{12}}\right] \approx \$8065.45$$

Find the monthly house payments necessary to amortize each loan. Then calculate the total payments and the total amount of interest paid.

11. \$175,000 at 6.24% for 30 years.

N:	$12 \cdot 30 = 360$
`I%:`	6.24
`PV:`	175,000
`PMT:`	?
`FV:`	0
`P/Y:`	12
`C/Y:`	12

Monthly payments: `PMT` $= -\ 1076.3671$ so payments of \$1076.37
Total payments: $1076.37 \cdot 360 = \$387,493.20$
Interet paid: $387,493.20 - 175,000 = \$212,493.20$

Formula Sheet

P =principal/present value r =annual interest rate m =compoundings per year $n = tm$
A =future/maturity value t =number of years $i = r/m$ =rate per period r_E =effective rate

Simple Interest	**Compound Interest**	**Continuous Compounding**
$A = P(1 + rt)$	$A = P(1 + i)^n$	$A = Pe^{rt}$
$P = \dfrac{A}{1 + rt}$	$P = A(1 + i)^{-n}$	$P = Ae^{-rt}$
	$r_E = \left(1 + \dfrac{r}{m}\right)^m - 1$	$r_E = e^r - 1$

R =periodic payment P =present value of annuity n =number of periods
S =future value i =rate per period

Future Value of an Annuity	**Sinking Fund Payment**
$S = R\left[\dfrac{(1+i)^n - 1}{i}\right]$	$R = \dfrac{S \cdot i}{(1+i)^n - 1}$

Present Value of an Annuity	**Amortization Payments**
$P = R\left[\dfrac{1 - (1+i)^{-n}}{i}\right]$	$R = \dfrac{P}{[1 - (1+i)^{-n}]}$

Ti-83/84

N : number of payments
`I%` : interest rate
`PV` : present value
`PMT` : payment size
`FV` : future value
`P/Y` : payments per year
`C/Y` : compoundings per year

\